Dr. Shuang Song

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EDUCATION BACKGROUND

Ph.D. in Physical Geography *01/09/2018 - 30/06/2023*
Beijing Normal University, Beijing, China, 100875.

Study of History (Double Major) *01/09/2014 - 30/06/2018*
Sun Yat-sen University, Guangdong, China, 510275.

B.S. of Science, Physical Geography *01/09/2014 - 30/06/2018*
Sun Yat-sen University, Guangdong, China, 510275.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher *01/01/2026 - present*
Department of Co-evolution of Land Use and Urbanisation, Max Planck Institute of Geoanthropology, Jena, Germany, 07745.

Postdoctoral Researcher *01/10/2024 - 31/12/2025*
Department of Co-evolution of Land Use and Urbanisation & Department Structural Changes of the Technosphere, Max Planck Institute of Geoanthropology, Jena, Germany, 07745.

Postdoctoral Researcher *01/07/2023 - 31/03/2025*
School of Environment & RDC for Watershed Environmental Eco-Engineering, Beijing Normal University, Beijing, China, 100875.

Doctoral Researcher *07/10/2017 - 30/06/2023*
State Key Laboratory of Earth Surface Processes and Resource Ecology, Faculty of Geography, Beijing Normal University, Beijing, China, 100875.

FUNDED SCIENTIFIC PROJECTS

[Co-PI] **Water - food - energy - ecosystem nexus in the Yellow River Basin from the perspective of reservoir operation**

*NSFC General Fund (Grantee: Xutong Wu, **Shuang Song**, 450,000 CNY)* *2025-2028*

This project aims to reveal the water-food-energy-ecosystem nexus in the Yellow River Basin, centering on the chain of ”ecological restoration - water and sediment changes - reservoir operations - downstream food production”, and to explore optimized operation strategies for the Xiaolangdi Reservoir under various scenarios that balance the needs for sediment reduction, hydropower generation, and irrigation. The findings will provide new insights into the water-food-energy-ecosystem nexus and offer a scientific basis for ecological conservation and high-quality development of the Yellow River Basin, thereby holding significant scientific and practical value.

[Principal Investigator] **Simulation and Optimization of Agricultural Irrigation Water Use in the Yellow River Basin by Agent-based Modeling**

*NSFC Young Scientists Fund (Grantee: **Shuang Song**, 300,000 CNY)* *2024-2027*

This project focuses on developing an agent-based model to simulate agricultural irrigation under various climate and policy scenarios. By integrating agent adaptability with public goods theory, we aim to identify the key drivers of changes in water use and predict future irrigation demand in response to climate shifts. This work builds on my previous experience with human-environment system modelling, directly contributing to future water management strategies in the river basin.

[Participate] **Human Management of the Water Cycle in a Global Comparative Perspective**

Joint Funded by Max-Planck Society (Grantee: Jürgen Renn, 200,000 ERU) *2024-2025*

The research plan operates across three integrated focus areas: Geoanthropology, real-time data integration, and water scarcity. This project aims to establish a global network of partners for global comparative studies. In this project, I collect water resource management case studies from around the world, create data visualisations, and network international partners.

[Participate] **Coordination mechanism and regulation strategy of multiple processes of water-sediment-ecology-economy system in the Yellow River Basin**

NSFC Joint Funds & Yellow River Water Science Funds (Grantee: Enhui Jiang, 11,720,000 CNY) 2023-2026

As a key participant in this interdisciplinary project, I contributed to the development of a coordinated strategy to manage the interconnected water-sediment-ecology-economy system of the Yellow River Basin. My role included research design, data collection, data analysis, and report drafting. The project explores the feedback mechanisms between environmental changes and socio-economic dynamics, with implications for long-term sustainability.

[Participate] **Mechanisms of human-natural system coupling and optimization of the Yellow River Basin**

NSFC National Major Research Plan (Grantee: Bojie Fu, 17,000,000 CNY) 2021-2025

This large-scale project investigates the coupling mechanisms between human and natural systems, focusing on optimization strategies for sustainable development in the Yellow River Basin. My contributions spanned data analysis, report writing, and project coordination. The research outcomes have been instrumental in improving socio-hydrological models and enhancing our understanding of long-term water resource management.

[Participate] **Sustainability of the socio-hydrological system**

NSFC Excellent Young Scientists Fund (Grantee: Shuai Wang, 1,300,000 CNY) 2017-2020

This project aimed to improve mathematical models of socio-hydrology by examining the dynamic interactions between human activities and water systems in the Yellow River Basin. I was involved in the project's theoretical and empirical aspects, including data collection, model development, and the publication of findings. This experience directly supports my current research on the long-term resilience of human-environment systems.

PUBLICATIONS

Dr. Shuang Song has published in the leading journals, such as *Nature Communications*, *Water Resources Research*, *Journal of Hydrology*, *Hydrology and Earth System Science*, and *Nature Sustainability*. Listed below are the 11 published or accepted articles with Dr. Song as **first or corresponding author**; for the complete list, including manuscripts under review, please visit https://cv.songshgeo.com/publication/.

[1] **Song, S.***, Hu, B., Findley, D. M., Wu, X., Liu, W., et al. **in press**. "Archival and Palaeoenvironmental Documentation of Extreme Climate Events Reveals Generational Amnesia in Collective Memory". In: *Nature Communications*.

[2] Hu, B., **Song, S.***, Wang, Y., Zhang, X., Liu, Y., et al. **2026a**. "Memory Sources Shape Flood Risk Perception and Agricultural Decision Among Farmers in a Drought-Oriented Irrigation District". In: *Journal of Flood Risk Management* 19.3, e70244. DOI: 10.1111/jfr3.70244.

[3] **Song, S.***, Zhang, L., Renn, J., and Roberts, P. **2026d**. "From Regime Shifts to Adaptive Cycles: Dynamic Understanding of Human–Water System". In: *Environmental Research: Water* 2.2. DOI: 10.1088/3033-4942/ae609a.

[4] **Song, S.**, Wang, S.*, Jiao, C., and Ibarra, E. J. M. **2024b**. "ABSESpy: An Agent-Based Modeling Framework for Social-Ecological Systems". In: *Journal of Open Source Software* 9.96, p. 6298. DOI: 10.21105/joss.06298.

[5] **Song, S.**, Wen, H., Wang, S., Wu, X., Cumming, G. S., et al. **2024d**. "Quantifying the Effects of Institutional Shifts on Water Governance in the Yellow River Basin: A Social-Ecological System Perspective". In: *Journal of Hydrology* 629, p. 130638. DOI: 10.1016/j.jhydrol.2024.130638.

[6] **Song, S.**, Wang, S.*, Wu, X., Wei, Y., Cumming, G. S., et al. **2023c**. "Identifying Regime Transitions for Water Governance in the Yellow River Basin, China". In: *Water Resources Research* 59.12, e2022WR033819. DOI: 10.1029/2022WR033819.

[7] **Song, S.**, Wang, S.*, Wu, X., Huang, Y., and Fu, B. **2022b**. "Decreased Virtual Water Outflows from the Yellow River Basin Are Increasingly Critical to China". In: *Hydrology and Earth System Sciences* 26.8, pp. 2035–2044. DOI: 10.5194/hess-26-2035-2022.

[8] **Song, S.**, Du, J.*, Wu, Q., Ni, M., Wang, Y., et al. **2021c**. "The Responses of *Spinifex Littoreus* to Sand Burial on the Coastal Area of Pingtan Island, Fujian Province, South China". In: *Écoscience*, pp. 1–10. DOI: 10.1080/11956860.2021.1888523.

[9] **Song, S.**, Wang, S.*, Fu, B., Dong, Y., Liu, Y., et al. **2021d**. "Improving Representation of Collective Memory in Socio-Hydrological Models and New Insights into Flood Risk Management". In: *Journal of Flood Risk Management* 14.1. DOI: 10.1111/jfr3.12679.

[10] **Song, S.**, Wang, S.*, Fu, B., Liu, Y., Wang, K., et al. **2020**. "Sediment Transport under Increasing Anthropogenic Stress: Regime Shifts within the Yellow River, China". In: *Ambio* 49.12, pp. 2015–2025. DOI: 10.1007/s13280-020-01350-8.

[11] **Song, S.**, Wang, S.*, Fu, B., Chen, H., Liu, Y., et al. **2019a**. "Study on adaptive governance of social-ecological system: Progress and prospect". In: *Acta Geographica Sinica* 74.11, pp. 2401–2410.

* = Corresponding author

TEACHING EXPERIENCE

[Lecturer] **GIS and Modelling Approaches**
IMPRS ModA, Max Planck Institute of Geoanthropology, Jena, Germany *10/2026*

Full module, taught independently.

[Guest Lecture] **Big Data and Computational Methods in Archaeology**
Department of Archaeology, Friedrich Schiller University Jena, Jena, Germany *04/2026*

One session, delivered on behalf of Prof. Patrick Roberts.

[Teaching Assistant] **Landscape Ecology, a graduate course of about 50 students**
Faculty of Geography, Beijing Normal University, Beijing, China *09/2019-01/2020*

STUDENT CO-SUPERVISION

Bo Hu *PhD Candidate*
Social Psychology, Nanjing University

Chentai Jiao *PhD Student*
Physical Geography, Beijing Normal University

Yueman Hou *PhD Student*
Physical Geography, Beijing Normal University

Shihao Li *Master's Student*
Archaeology, Shandong University

Jiayi Que *Master's Student*
Historical Geography, Renmin University of China

OUTREACH PROJECT EXPERIENCE

[Founder] **ABMind: a Chinese-language learning community for agent-based modelling**
Community website: https://abmind.cc/ *10/2024-present*

I founded and run this community to make agent-based modelling approachable to Chinese-speaking researchers and students, who are otherwise served almost entirely by English-language material.

[Co-PI] **Ecosystem services in Inner Mongolia**
*Inner Mongolia Key Scientific Project (Grantee: **Joint sources**)* *2021-present*

In this project, we examine Inner Mongolian grassland communities to explain the dangers posed by ecological degradation. I have gone deep into the grasslands four times to conduct interviews and research with local herders to understand ecological issues.

[Personal Project] **Popular Blogs: A geographer who also travels**
Monthly publish articles on https://songshgeo.com/ *2018-present*

In this series of writings, I position myself as "A Geographer who also travels." I write in Chinese, combining geographic knowledge, academic concepts, and travel experiences, and convey them to the public in an easy-to-understand language. I have earned 4,688 CNY tips for this blog, with 812 subscriptions and readership. The webpage has around 4,000 visitors.

PROFESSIONAL PRESENTATIONS

[Invited Presentation] **Societal Adaptation to Environmental Dynamics: Model Simulation Perspective.**
Social Psychology Laboratory, Arizona State University, Tempe, USA *12/2026*

[Oral Presentation] **Coupling flood and agent-based models for interpreting centuries of river-urbanism interactions.**
2026 International Conference on Natural Resources and Planning, Hong Kong, China *28/03/2026*

[Oral Presentation] **Coupling flood and agent-based models for interpreting centuries of river-urbanism interactions.**
On the Way to the Fluvial Anthroposphere (DFG SPP 2361), Tübingen University, Tübingen, Germany *21/01/2026*

[Oral Presentation] **Millennial-scale data reveals a collective memory of extreme drought and flood events.**
University of Tokyo, Tokyo, Japan *20/07/2025*

[Invited Presentation] **Research on the Long-term Co-evolution of Human Ecology and Water Environment Based on Model Simulation.**
Shanxi Normal University, Xi'an, China *31/03/2025*

[Invited Presentation] **Co-evolution between Hydro-system and Historical cities.**
Harbin Institute of Technology (Shenzhen Campus), Shenzhen, China *17/03/2025*

[Oral Presentation] **The Evolution Process and Driving Mechanisms of the Human-Water Relationship in the Yellow River from the Perspective of Watershed Governance.**
Ningxia University, Ningxia, China *19/07/2024*

[Oral Presentation] **The Evolution Process and Driving Mechanisms of the Human-Water Relationship in the Yellow River from the Perspective of Water Governance. (In Chinese)**
1st Hydrological Process Forum, Shenzhen, China *26/04/2024*

[Oral Presentation] **Empowering Human-Water System Analysis through ABSESpy: An Agent-Based Modeling Framework of SES.**
European Geosciences Union Assembly 2024, Vienna, Austria *19/04/2024*

[Oral Presentation] **Research on a coupling model of Human-Earth system -A case study of agent-based model**
EGU General Assembly 2023, Vienna, Austria *25/04/2023*

[Oral Presentation] **Analysis of the Complexity of Virtual Water Transfer in the Yellow River Basin**
Academic Forum on Ecological Civilization and Sustainable Development, China *22/11/2021*

[Oral Presentation] **Study on human-water relationship change in the Yellow River Basin**
Conference of State Key Laboratory of Earth Surface Processes and Resource Ecology, Beijing, China *19/06/2021*

[Oral Presentation] **Sediment Transport under Increasing Anthropogenic Stress: Regime Shifts Within the Yellow River, China.**
AGU Fall Meeting 2019, San Francisco, USA *10/12/2019*

HONORS AND AWARDS

Academic Innovation Award at Beijing Normal University	*05/2025*
Beijing Excellent Doctoral Thesis	*07/2024*
Excellent Doctoral Thesis (Top 1 in Geography)	*04/2024*
2nd in the 8th National Disaster Reduction and Emergency Management Academic Competition	*05/2020*
1st & Best Lecturer in the Third Academic Speech Competition of Beijing Normal University	*05/2019*

1st in the 10th Experimental Science Championship of Systems Science *11/2018*

Outstanding graduate thesis (Top 1 in Physical Geography) *06/2018*

Outstanding Project in National College Student Innovation Funding *04/2017*

1st & Best Lecturer in the 3rd China University Geography Science Presentation Competition *09/2017*

1st & Most Novel Topic in the 2nd China University Geography Science Presentation Competition *11/2016*

GRANTS AND SCHOLARSHIPS

Seal of Excellence (95.6 points), MSCA Action, Horizon Europe *02/2026*

Visiting Scholarship from Max-Planck Society (15,000 EUR) *10/2024 - 03/2025*

Young Scientist Fund of NSFC (300,000 CNY) *08/2024*

Beijing Normal University Scholarships (Four-time 1st & once 2nd prize, 58,000 CNY) *12/2018 - 12/2022*

Graduate Student Academic Ability Competition (Twice 1st & Once 2nd, 8,000 CNY) *12/2018 - 12/2022*

Chinese National Scholarship (Top 5%, 20,000 CNY) *12/2020*

Sun Yat-sen University Scholarships. (Twice 1st, 10,000 CNY) *12/2014 - 12/2018*

"1987" Economic Geography Scholarship (1,500 CNY) *12/2018*

Guanghua Education Scholarship (3,000 CNY) *12/2017*

Chinese National Scholarship (Top 5%, 8,000 CNY) *11/2016*

National College Student Innovation Funding (10,000 CNY, Excellent enclosed) *11/2016*

ACADEMIC SERVICES

Reviewer for Academic Journals:

Peer reviewer for **34** journals (**43** reviews since 2019) across **5** fields; **26** of them are top-quartile (Q1) in their JCR category.

Water Resources & Hydrology *Journal of Hydrology: Regional Studies* (Q1, 2026) · *Groundwater for Sustainable Development* (Q1, 2026) · *Water Resources Research* (Q1, 2026) · *Frontiers in Water* (Q2, 2026) · *Agricultural Water Management* (Q1, 2025) · *Frontier of Freshwater* (2025) · *Journal of Hydrology* (Q1, 2024) · *Hydrology and Earth System Sciences* (Q1, 2024) · *Water Research* (Q1, 2021)

Sustainability & Human-Environment Systems *World Development Sustainability* (Q1, 2026) · *Geography and Sustainability* (Q1, 2026) · *Energy Nexus* (Q1, 2026) · *Sustainable Cities and Society* (Q1, 2026) · *Ecological Economics* (Q1, 2026) · *AMBIO: A Journal of the Human Environment* (Q1, 2024) · *People and Nature* (Q1, 2024) · *Journal of Cleaner Production* (Q1, 2019)

Environmental Policy & Governance *Environmental Science & Policy* (Q1, 2026) · *Earth System Governance* (Q1, 2025) · *Regional Science Policy & Practice* (Q2, 2025) · *Utilities Policy* (Q2, 2025) · *Progress in Disaster Science* (Q1, 2025) · *Journal of Environmental Management* (Q1, 2023)

Earth & Environmental Change *Geomorphology* (Q2, 2026) · *Journal of Environmental Sciences* (Q1, 2026) · *Physics and Chemistry of the Earth* (Q1, 2026) · *Regional Ecology and Management* (2025) · *Journal of Arid Environments* (Q2, 2025) · *Global and Planetary Change* (Q1, 2025) · *Quaternary International* (Q2, 2024)

Multidisciplinary & Modelling *Environmental Modelling & Software* (Q1, 2026) · *iScience* (Q1, 2026) · *Scientific Reports* (Q1, 2026) · *Nature Communications* (Q1, 2025)

Professional consultations for:

- *Urat Center banner Grassland Ecological Research Association* *03/2024-Present*
- *Yellow River Institute of Hydraulic Research, Yellow River Conservancy Commission* *08/2022*

Teaching experience:

- *Extended Teaching Faculty International Max Planck Research School (IMPRS ModA).* *12/2025-Present*
- *Agent-based model Seminar Series Teaching Faculty at MPI-GEA* *09/2025-Present*

OUTREACH SPEECH

[Lecture] **Water Management in Decision Theater - Learning from Others. (Chinese Case)**
Falling Walls Science Summit, Berlin, Germany *08/11/2025*

[Lecture] **Research on coupling model of Human-Earth system - A case study of agent-based model**
Graduate course, Faculty of Geography, Beijing Normal University, Beijing, China *23/04/2024*

[Invited Presentation] **Coupled human and water in the Anthropocene: Social-ecological system (SES) modeling approach**
Annual Academic Conference of Faculty of Geography, Beijing Normal University, Beijing, China *24/12/2023*

[Invited Lecture] **ABSESpy, Agent-Based Social-ecological systems Modelling Framework in Python**
Sun Yat-sen University, Guangzhou, China *20/11/2023*

[Invited Lecture] **Coupled human and water in the Anthropocene: Social-ecological system (SES) modeling approach**
Universidad San Francisco de Quito, Quito, Ecuador *15/09/2023*

[Invited Lecture] **Experience sharing in scientific data visualization**
Yellow River Conservancy Research Institute, Zhengzhou, China *17/07/2023*

[Seminar] **Evolution and driving mechanisms of human-water relationship in the Yellow River from the perspective of basin governance**
Graduate course, Faculty of Geography, Beijing Normal University, Beijing, China *23/04/2023*

[Seminar] **Flooding Collective Memory Resilience**
International Institute for Applied Systems Analysis: IIASA, Vienna, Austria *04/04/2023*

[Invited Lecture] **Look for interdisciplinary imagination**
Outstanding Student Research Experience Sharing Conference, Beijing Normal University *25/05/2022*

[Invited Lecture] **Research Methods for Complex System Management: Application of Agent-based Modeling in Groundwater Resource Management**
Virtual Lecture at Seminar of Wisdom Lab, 100+ Audiences, China *22/12/2021*

[Invited Lecture] **Interdisciplinary and Problem-Oriented Sustainability Science**
Renmin University of China, Beijing, China *22/05/2021*

SOFTWARE DEVELOPED

[Leading developer] **ABSESpy: An agent-based modeling framework for social-ecological systems**
Open-sourced software on GitHub: https://github.com/ABSESpy/ABSESpy *23/12/2022-Present*

It is based on the popular Python language and the general Mesa framework, enabling it to simulate complex interactions between agents and their environment. It has significantly improved in supporting real-world simulations.

[Team Leader] **PaperBell: Research, to be connected.**
Open-sourced software on GitHub: https://github.com/PaperBell-Org/Obsidian-PaperBell *2024-present*

I developed this ecosystem that provides researchers with an automatic academic note-taking workflow. This open-source project currently has 216 stars on GitHub and an active community.

OTHER EXPLORATIONS

[Participated] **Zambezi River Basin Water Resources Management Survey**
NSFC funded exploration *06/04/2025-15/04/2025*

[Organized] **Andes Mountains, Amazon Rainforest, and Galápagos Islands**
Personal exploration *08/2023-10/2023*

[Participated] **The Second Chinese Scientific Expedition to the Qinghai-Tibet Plateau**
National funded exploration *07/2021-09/2021*

[Participated] **Investigation of Grassland Ecological Degradation in Inner Mongolia**
Inner Mongolia Government funded exploration *Four times from 07/2020 to 09/2022*